IGCSE 0607 Practice Questions — E6.2 — Combined Questions & Markscheme

Area and perimeter

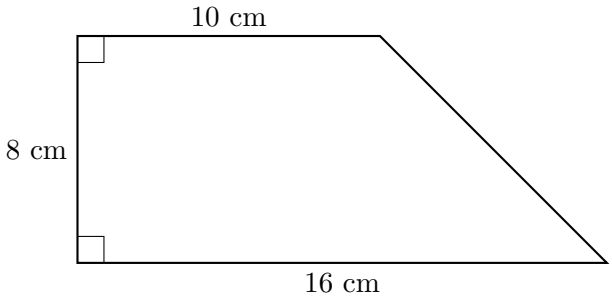
Total Questions: 6 — **Total Marks:** 41 Marks

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  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

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1 The diagram shows a right-angled trapezium.



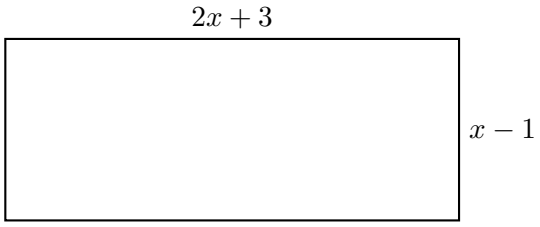
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[3]**

2 The diagram shows a rectangle.
All measurements are in centimetres.



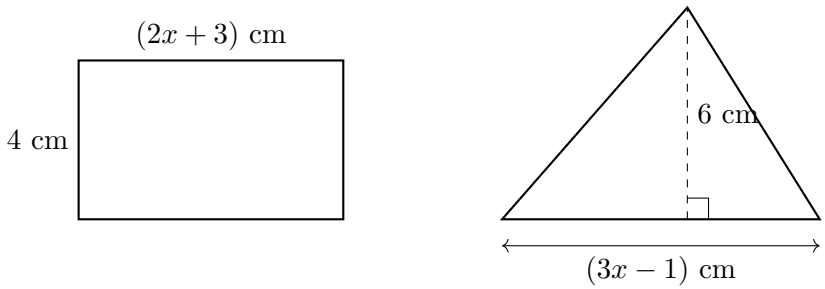
(a) The perimeter of the rectangle is 34 cm.
Find the value of x.

$x =$ ..... **[3]**

(b) Calculate the area of the rectangle.

..... cm^2 **[2]**

3 The diagram shows a rectangle and a triangle.
The area of the rectangle is equal to the area of the triangle.



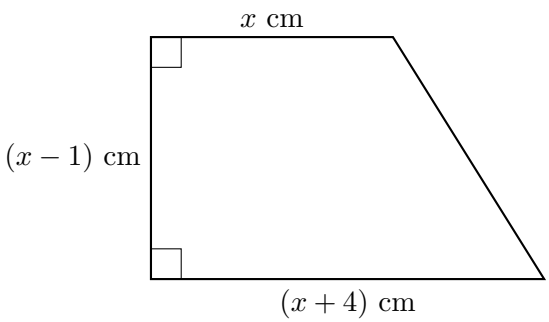
(a) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[4]**

(b) Work out the perimeter of the rectangle.

..... cm **[2]**

4 The diagram shows a right-angled trapezium with parallel sides of length x cm and $(x + 4)$ cm.
The height of the trapezium is $(x - 1)$ cm.
The area of the trapezium is 40 cm^2.



(a) Show that $x^2 + x - 42 = 0$.

[3]

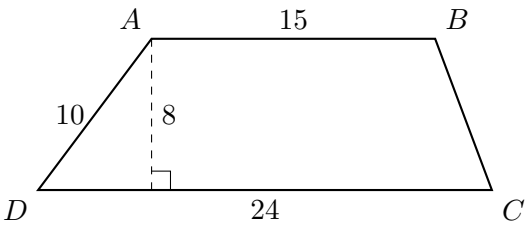
(b) Solve the equation $x^2 + x - 42 = 0$ to find the value of x.

$x =$ ..... **[2]**

(c) Calculate the perimeter of the trapezium.

..... cm **[4]**

5 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The perpendicular height of the trapezium is 8 cm.
All measurements are in centimetres.



(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[4]**

6 A piece of wire of length 60 cm is cut into two pieces.
One piece is bent to form a square.
The other piece is bent to form a rectangle where the length is exactly twice the width.
Let the width of the rectangle be y cm.

(a) Write down an expression, in terms of y, for the side length of the square.

..... cm **[2]**

(b) The total area enclosed by the square and the rectangle is 113 cm^2.
Show that $17y^2 - 180y + 448 = 0$.

[3]

(c) Solve the equation.

$$17y^2 - 180y + 448 = 0$$

Show all your working.

$y =$ ..... , $y =$ ..... **[3]**

(d) Given that the side length of the square must be greater than the width of the rectangle, calculate the area of the square.

..... cm^2 **[2]**

1 **(a)** $\frac{1}{2}(10+16)\times 8$ **[M1]**

104 **[A1]**

(b) Base of right-angled triangle $= 16-10=6$ **[M1]**

Slanted edge $=\sqrt{6^2+8^2}=10$ **[M1]**

$8+10+16+10=44$ **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** $2(2x+3)+2(x-1)=34$ OR $2x+3+x-1=17$ **[M1]**

$6x+4=34$ OR $3x+2=17$ **[M1]**

$x=5$ **[A1]**

(b) Substitution of $x=5$: $(2(5)+3)\times(5-1)$ OR 13×4 **[M1]**

52 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** Area of rectangle $=4(2x+3)$ OR $8x+12$ **[M1]**

Area of triangle $=\frac{1}{2}\times 6\times(3x-1)$ OR $9x-3$ **[M1]**

$8x+12=9x-3$ **[M1]**

$x=15$ **[A1]**

(b) Length $=2(15)+3=33$ **[M1]**

Perimeter $=2(4+33)=74$ **[A1]**

[Back to Question] *[Total Marks: 6]*

4 **(a)** $\frac{1}{2}(x+x+4)(x-1)=40$ **[M1]**

$\frac{1}{2}(2x+4)(x-1)=40$ OR $(x+2)(x-1)=40$ **[M1]**

$x^2+2x-x-2=40 \implies x^2+x-42=0$ **[A1]**

(b) $(x+7)(x-6)=0$ **[M1]**

6 **[A1]**

(c) Identifies lengths are Top $=6$, Bottom $=10$, Height $=5$ **[M1]**

Horizontal difference $=4$, so slanted side $=\sqrt{5^2+4^2}$ OR $\sqrt{41}$ **[M1]**

Perimeter $=5+6+10+\sqrt{41}$ **[M1]**

27.4 (or $27.40\ldots$) **[A1]**

[Back to Question] *[Total Marks: 9]*

5 **(a)** $\frac{1}{2} \times (15 + 24) \times 8$ **[M1]**

156 **[A1]**

(b) Horizontal distance from D to the altitude from $A = \sqrt{10^2 - 8^2} = 6$ **[M1]**

Horizontal distance for the right-angled triangle under $B = 24 - 15 - 6 = 3$ **[M1]**

$BC = \sqrt{8^2 + 3^2}$ $(= \sqrt{73} \approx 8.544)$ **[M1]**

57.5 OR 57.54 to 57.55 **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** Perimeter of rectangle $= 6y$ OR Perimeter of square $= 60 - 6y$ **[M1]**

$\frac{60 - 6y}{4}$ OR $15 - 1.5y$ **[A1]**

(b) $(15 - 1.5y)^2 + y(2y) = 113$ **[M1]**

$225 - 45y + 2.25y^2 + 2y^2 = 113$ **[M1]**

$4.25y^2 - 45y + 112 = 0$ leading to $17y^2 - 180y + 448 = 0$ with no errors seen **[A1]**

(c) $\frac{-(-180) \pm \sqrt{(-180)^2 - 4(17)(448)}}{2(17)}$ **[M1]**

$\frac{180 \pm \sqrt{1936}}{34}$ OR $\frac{180 \pm 44}{34}$ **[M1]**

$y = 4$ and $y = 6.59$ (or 6.588... or $\frac{112}{17}$) **[A1]**

(d) Identifies $y = 4$ as the valid solution (since $15 - 1.5(6.59) < 6.59$) leading to side length $= 9$ **[M1]**

81 **[A1]**

[Back to Question] *[Total Marks: 10]*